HSTNet: Hybrid Supervision-Driven Two-Stream Collaborative Network for Hyperspectral Wheat Variety Classification

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***Abstract*—Hyperspectral remote sensing plays an important role in agricultural monitoring, and fine-grained wheat variety classification is essential for advancing smart agriculture. However, progress is limited by the scarcity of high-quality spectral samples and the challenges associated with collecting large-scale hyperspectral data. To cope with these issues, we design a hybrid supervised-driven two-stream collaborative network (HSTNet), which consists of a semi-supervised conditional generative adversarial network (SCGAN) for data augmentation and a supervised two-stream discriminative network (STDNet) for wheat classification. In SCGAN, the generator constructs a mapping relationship between input noise and real wheat hyperspectral samples to generate fake wheat hyperspectral samples that are highly matched with the distribution of the real sample, and the discriminator with multilayer perceptions utilizes discriminative learning to identify real and fake samples. In STDNet, it collaboratively extracts the spectral, spatial, and texture features of wheat hyperspectral images employing the dual-stream branch structure of 3-D convolutional neural network (3DCNN) and 2-D convolutional neural network (2DCNN). Subsequently, it utilizes the fast Fourier transform (FFT) and the cross-attention mechanism (CAM) to refine and fuse these features to improve their capability of feature expression. Noteworthy, the individual design effectively improves the classification results of wheat varieties via collaborative optimization among modules. Besides, we built a mixed wheat hyperspectral dataset (MWHD) with 4800 samples of 20 wheat varieties. Extensive experiments on our constructed MWHD dataset demonstrate that the proposed HSTNet outperforms state-of-the-art methods in wheat variety classification. The code is publicly available at: https://github.com/bakam412/HSTNet**

***Index Terms*—Conditional generative adversarial, hybrid supervision, two-stream discriminative, wheat variety classification.**

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I. Introduction

With the rapid development of breeding technologies, the number of wheat hybrid varieties has increased significantly, accompanied by substantial differences in growth traits and disease resistance. Accurate variety classification is therefore crucial for precise field management, yield improvement, and quality optimization. However, the coexistence of mixed varieties, the large number of categories, and the high similarity of phenotypic characteristics present major challenges for modern smart agriculture [1]. Hyperspectral imaging has gained widespread use in agricultural monitoring due to its ability to capture fine, nanoscale spectral signatures of crops [2]. Compared with traditional chemical analysis, deep learning combined with hyperspectral imaging enables more comprehensive and accurate variety identification [3], [4], providing a solid theoretical and technical foundation for crop optimization and yield enhancement, and offering new momentum for the sustainable development and modernization of smart agriculture.

In the data augmentation, prior studies have shown that increasing the diversity and richness of generated samples can effectively improve the performance and robustness of deep learning models across various visual tasks [5], [6], [7], [8], [9]. Despite this, current research on wheat variety classification still relies heavily on conventional augmentation techniques and incremental network optimization, which limits the achievable performance gains. In terms of data augmentation, traditional methods mainly include random cropping and flipping, channel swapping, and interpolation [10], [11], while deep learning mainly includes generative adversarial network (GAN) [12], [13] and variational autoencoder (VAE) [14]. These limitations highlight the need for

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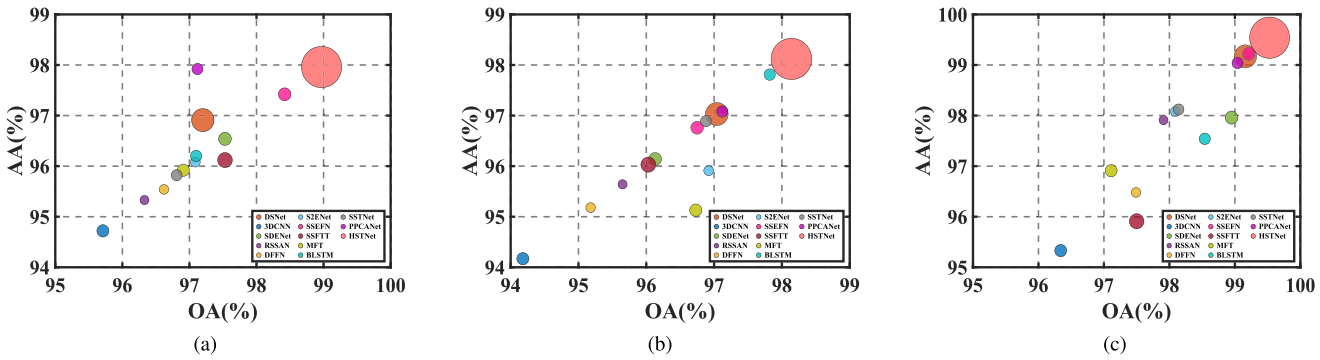


Fig. 1. Performance comparison of various methods on (a) real dataset, (b) generated dataset, and (c) mixed dataset. Notably, the horizontal and vertical axes represent OA and AA, respectively. The upper-right area of each plot reflects superior performance, and the bubble size indicates the model's accuracy.

more advanced augmentation strategies capable of producing structurally diverse, semantically faithful, and task-relevant samples. Although traditional augmentation methods are simple and easy to implement, they inherently fail to capture the complex spatial–spectral relationships of hyperspectral data and cannot accurately model real-world interference factors, resulting in limited improvements in downstream tasks. To overcome these shortcomings, VAEs have been introduced to generate hyperspectral samples by learning latent probability distributions [15], [16]. However, existing studies report that VAE-generated samples still suffer from notable deficiencies in quality and diversity, with insufficient sample variability being a persistent bottleneck [17]. GAN-based approaches, which employ adversarial optimization strategies, have also been explored for data augmentation. Nonetheless, they are prone to instability, mode collapse, and difficulties in generating spectrally consistent hyperspectral samples. These limitations collectively indicate that current augmentation methods remain inadequate for capturing the intrinsic complexity of hyperspectral data. Specifically, GANs attempt to capture the latent data distribution through adversarial training between generators and discriminators to produce samples that closely resemble real data [5], [18]. However, despite their strong generative capacity, GANs remain difficult to train and are highly susceptible to mode collapse, which limits their ability to generate diverse and spectrally consistent hyperspectral samples [19], [20]. To address these limitations, we integrate the complementary strengths of VAEs and GANs and propose a semi-supervised conditional GAN (SCGAN) that leverages spectral band means and variances to construct a more stable and informative generative process.

In hyperspectral image classification, mainstream approaches include machine learning-based methods [21], [22] and deep learning-based methods [23], [24], [25]. Machine learning techniques rely on hand-crafted or partially manual feature extraction, making them difficult to scale and less effective when dealing with high-dimensional spectral data or large, complex datasets. Deep learning alleviates this issue by automatically learning hierarchical features, and 2-D convolutional neural networks (2DCNNs) have therefore been increasingly adopted for hyperspectral classification [26]. However, despite their strong spatial feature extraction capability, 2DCNNs inherently underutilize the rich spectral information, leading to incomplete spectral–spatial feature representation and consequently limiting classification performance [27], [28]. These shortcomings indicate that existing methods still struggle to fully exploit the joint spectral–spatial characteristics essential for accurate hyperspectral image classification. Recently, hybrid models that combine 3-D convolutional neural networks (3DCNNs) with 2DCNNs have shown promise in capturing both spectral and spatial information in hyperspectral images [29], [30]. However, existing fusion strategies often rely on simple concatenation or parallel feature extraction, which limits their ability to fully model the complex spectral–spatial interactions and to effectively separate task-relevant information from redundant or correlated features. Moreover, many current designs overlook the frequency-domain characteristics of hyperspectral data, resulting in incomplete feature representation. Motivated by these limitations, we develop a supervised two-stream identification network for hyperspectral wheat variety classification, which employs the fast Fourier transform (FFT) and a cross-attention mechanism (CAM) to adaptively guide stream separation, while jointly integrating 3DCNN and 2DCNN to more comprehensively extract spectral, spatial, and textural features from hyperspectral wheat images.

The scarcity and high acquisition cost of hyperspectral data have limited the large-scale deployment of hyperspectral classification models in smart agriculture, motivating our focus on sample augmentation and variety identification. To address this, we propose a hybrid supervision-driven two-stream collaborative network for hyperspectral wheat classification, termed HSTNet, which integrates two subnetworks: SCGAN and STDNet. In SCGAN, a semi-supervised GAN is employed, where the generator and discriminator operate concurrently under a game-theoretic framework to synthesize high-fidelity and diverse hyperspectral wheat images. Meanwhile, STDNet further refines the expressiveness of features using an FFT and CAM. Notably, we utilize the generated samples to feed the classifier in real-time for learning and training. The generated samples are fed to the classifier in real time, enabling dynamic classification and establishing a unified framework for simultaneous data augmentation and classification via end-to-end optimization. This integrated strategy not only streamlines the traditional workflow but also provides

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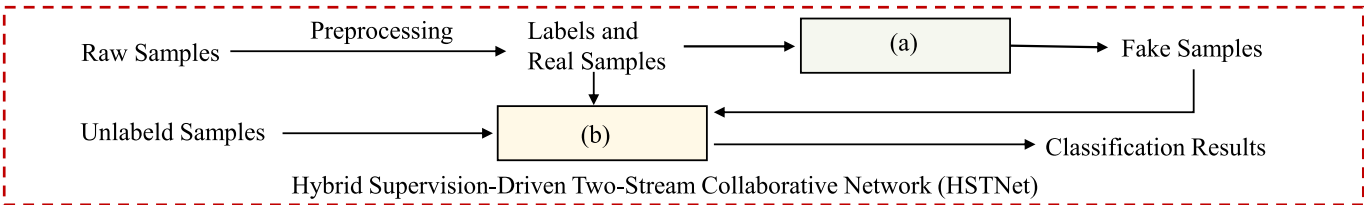


Fig. 2. Detailed workflow of HSTNet application in the hyperspectral wheat variety classification. In the HSTNet method, which consists of an SCGAN and an STDNet.

additional supervision for the GAN, effectively improving the overall performance of HSTNet. In addition, Fig. 1 presents a comparison of different methods on real, generated, and mixed datasets. Our HSTNet achieves the best balance between overall accuracy (OA) and average accuracy (AA) across all three datasets. As shown in Fig. 1, which reports results against 12 benchmark methods, HSTNet consistently exhibits the largest bubbles in all scenarios. Meanwhile, this indicates superior OA and AA, while also visually confirming the robustness and stability of our HSTNet in hyperspectral wheat variety classification. The distinct separation and dominance of HSTNet in the bubble chart further demonstrate the effectiveness of the proposed framework over existing methods.

The key contributions of our work are outlined as follows.

1) *Dataset Contribution:* We constructed a new mixed wheat hyperspectral dataset (MWHD) containing 4800 samples from 20 wheat varieties, including both real and SCGAN-generated samples. This dataset provides valuable benchmark data for advancing hyperspectral crop variety identification with deep learning.
2) *Framework Contribution:* We designed a collaborative framework that integrates adaptive preprocessing, data augmentation, and classification. The preprocessing stage employs a sliding window with adaptive weighting coefficients optimized iteratively to suppress noise and enhance spectral features, enabling SCGAN to model data distributions more effectively and supporting STDNet in learning discriminative spectral–spatial features.
3) *Methodological Contribution:* We proposed a hybrid supervision-driven two-stream collaborative network, HSTNet, which combines SCGAN and STDNet. SCGAN leverages a generator with long short-term memory (LSTM) and a discriminator with a multilayer perceptron to capture latent feature distributions, while STDNet integrates FFT with 3D-CNN and 2D-CNN under mixed supervision to achieve robust and accurate classification of hyperspectral wheat images.

The rest of this article is organized as follows. Section II reviews related work, Section III details the proposed HSTNet, Section IV presents the experimental results, and Section V concludes with a discussion of future research directions.

II. Related Work

This section presents an in-depth review of related studies on data augmentation and hyperspectral wheat variety classification.

A. Data Preprocessing

Hyperspectral imagery provides rich spectral information that captures fine physical and chemical properties of crops, yet its high dimensionality and feature redundancy pose major challenges for classification. Early studies primarily addressed this through dimensionality reduction; for instance, Xia et al. [31] and Chen et al. [32] applied principal component analysis (PCA) to compress hyperspectral data into lower dimensional subspaces while preserving essential information. Building on this, Sun et al. [33] segmented spectra for pixel-level labeling, improving feature discrimination and classification accuracy. Despite these advances, hyperspectral data remain sensitive to external factors such as illumination, atmospheric conditions, and terrain, which complicates feature extraction. To reduce such variability, Hong et al. [34] proposed a data-driven unmixing approach. Collectively, these studies highlight the value of spectral information while emphasizing that effective preprocessing is essential to enhance robustness and enable reliable fine-grained classification of wheat varieties in practical agricultural applications.

B. Data Augmentation

High-quality and diverse training data are crucial for deep learning models to achieve robust performance. Traditional augmentation methods, such as sample repetition [35], region-of-interest stacking [36], and geometric transformations [10], can increase dataset size but do not capture the intrinsic spectral–spatial structure of hyperspectral data. Recently, VAEs have been employed to generate hyperspectral images due to their capability for data synthesis and representation learning. For example, Wei et al. [37] proposed a graph-regularized VAE that leverages superpixel segmentation to model interpixel relationships and reconstruct spectral vectors, effectively augmenting data but substantially increasing computational complexity. To improve efficiency, Tian et al. [38] designed an orthogonal subspace bootstrapping VAE to model background distributions and mitigate the impact of limited labeled samples. Despite these advances, VAEs remain constrained by reconstruction loss and regularization, often producing samples with limited feature fidelity. These observations highlight the need for more advanced augmentation strategies to alleviate data scarcity, enhance generalization, and improve classification performance in hyperspectral wheat imagery.

Unlike VAEs, GANs can generate more realistic and detailed samples due to their adversarial training mechanism. For instance, Zhang et al. [39] proposed an attention-guided

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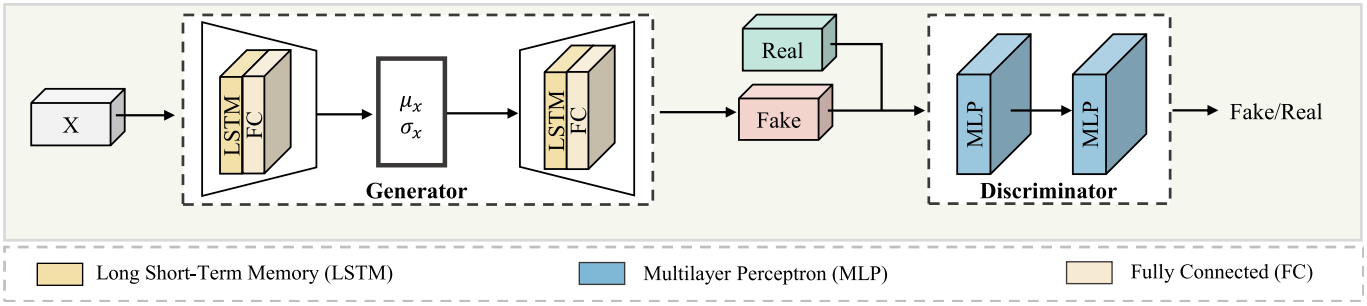


Fig. 3. Detailed workflow of SCGAN. In the SCGAN, which includes a generator and a discriminator. Our SCGAN fully exploits the distribution characteristics between noise and real samples via semi-supervised learning of generators and discriminators to generate samples that are as close as possible to real samples.

GAN to mitigate the scarcity of labeled samples caused by limited hardware resources. Jiang et al. [40] introduced a hyperspectral anomaly detection GAN that exploits unsupervised learning to identify and remove abnormal spectra, improving sample quality. Zhang et al. [41] designed a generative adversary-driven cross-perception network that fuses 2-D and 3-D convolutions for both data augmentation and wheat classification. Feng et al. [42] presented a self-supervised partitioned GAN to address spectral category imbalance, leveraging representation learning from clustering to enhance data balance and classification performance. Collectively, these studies demonstrate the strong potential of GANs for hyperspectral data augmentation, improving both the robustness and accuracy of classification models. Motivated by this, we propose SCGAN, a semi-supervised generative framework specifically designed to produce high-fidelity, diverse wheat hyperspectral samples, thereby enriching training data and enabling more effective classification.

C. Hyperspectral Image Classification

Currently, hyperspectral image classification mainly utilizes machine learning and deep learning approaches. In the following, we offer a detailed review of these two primary categories of classification techniques.

In traditional machine learning, the performance of hyperspectral image classification heavily depends on manual feature engineering. Techniques such as support vector machines (SVMs) [21], random forests (RFs) [43], and sparse representation [44] classify hyperspectral images by mapping pre-extracted features to category labels. Sheykhmousa et al. [45] systematically compared SVM and RF across multiple public spectral datasets, showing that SVM excels in handling high-dimensional spectral features by maximizing the margin in feature space, whereas RF is more effective in multisource data fusion by leveraging ensemble learning to capture diverse feature representations. To address these limitations, Tu et al. [46] proposed a weighted nearest-neighbor joint sparse representation method, which introduces a Gaussian weighting function to compute weighted Euclidean distances, mitigating the equal-weight constraint of traditional KNN and enhancing spatial information utilization. Overall, the feature extraction capability of traditional machine learning remains limited when confronting the inherently high-dimensional nature of hyperspectral data.

TABLE I
MODEL STRUCTURE AND PARAMETER DETAILS

Layer (type:depth-idx)	Input Shape	Output Shape	Param #
HSTNet	[1, 1, 30, 28, 28]	[1, 9]	5,547,216
Sequential: 1-1	–	–	–
Conv3d:	[1, 1, 30, 28, 28]	[1, 16, 24, 28, 28]	1,024
ReLU:	[1, 16, 24, 28, 28]	[1, 16, 24, 28, 28]	–
SpatialAttention: 1-2	[1, 384, 28, 28]	[1, 1, 28, 28]	–
Conv2d:	[1, 2, 28, 28]	[1, 1, 28, 28]	98
Sigmoid:	[1, 1, 28, 28]	[1, 1, 28, 28]	–
Sequential: 1-3	[1, 30, 28, 28]	[1, 64, 28, 28]	–
Conv2d:	[1, 30, 28, 28]	[1, 64, 28, 28]	17,344
ReLU:	[1, 64, 28, 28]	[1, 64, 28, 28]	–
ChannelAttention: 1-4	[1, 64, 28, 28]	[1, 64, 1, 1]	–
Conv2d:	[1, 64, 1, 1]	[1, 4, 1, 1]	256
ReLU:	[1, 4, 1, 1]	[1, 4, 1, 1]	–
Conv2d:	[1, 4, 1, 1]	[1, 64, 1, 1]	256
FrequencyMLP: 1-5	[1, 384, 28, 28]	[1, 256]	–
Sequential:	[1, 602112]	[1, 256]	–
Linear:	[1, 602112]	[1, 512]	308,281,856
LayerNorm:	[1, 512]	[1, 512]	1,024
GELU:	[1, 512]	[1, 512]	–
FrequencyMLP: 1-6	[1, 64, 28, 28]	[1, 256]	–
Sequential:	[1, 100352]	[1, 256]	–
Linear:	[1, 100352]	[1, 512]	51,380,736
LayerNorm:	[1, 512]	[1, 512]	1,024
GELU:	[1, 512]	[1, 512]	–
Sequential: 1-7	[1, 512]	[1, 128]	–
Linear:	[1, 512]	[1, 256]	131,328
LayerNorm:	[1, 256]	[1, 256]	512
Sequential: 1-8	[1, 576, 28, 28]	[1, 368, 28, 28]	–
Conv2d:	[1, 576, 28, 28]	[1, 368, 28, 28]	1,908,080
BatchNorm2d:	[1, 368, 28, 28]	[1, 368, 28, 28]	736
ReLU:	[1, 368, 28, 28]	[1, 368, 28, 28]	–
Linear: 1-9	[1, 288512]	[1, 128]	36,929,664
Dropout: 1-10	[1, 128]	[1, 128]	–
Linear: 1-11	[1, 128]	[1, 9]	1,161

Deep learning employs an end-to-end training paradigm, enabling automated feature extraction with greater generalization and robustness than traditional methods [47], [48]. For example, Zhong et al. [49] used 3-D convolutions to build a spectral–spatial residual network that captures both spectral and spatial features, integrating residual connections to enhance performance. Zhang et al. [41] proposed a generative adversarial-driven cross-aware network combining 2-D and 3-D convolutions for wheat variety recognition, supporting both data augmentation and classification. Zhu et al. [50] developed a residual spectral–spatial attention network, demonstrating the effectiveness of attention in modeling spectral–spatial correlations, while Sun et al. [4] introduced a spectral–spatial feature tokenization transformer that learns high-level semantic representations from convolutional features to improve classification accuracy. Building on these advances, we propose STDNet, a spectral–temporal dual-stream network for hyperspectral wheat variety classification.

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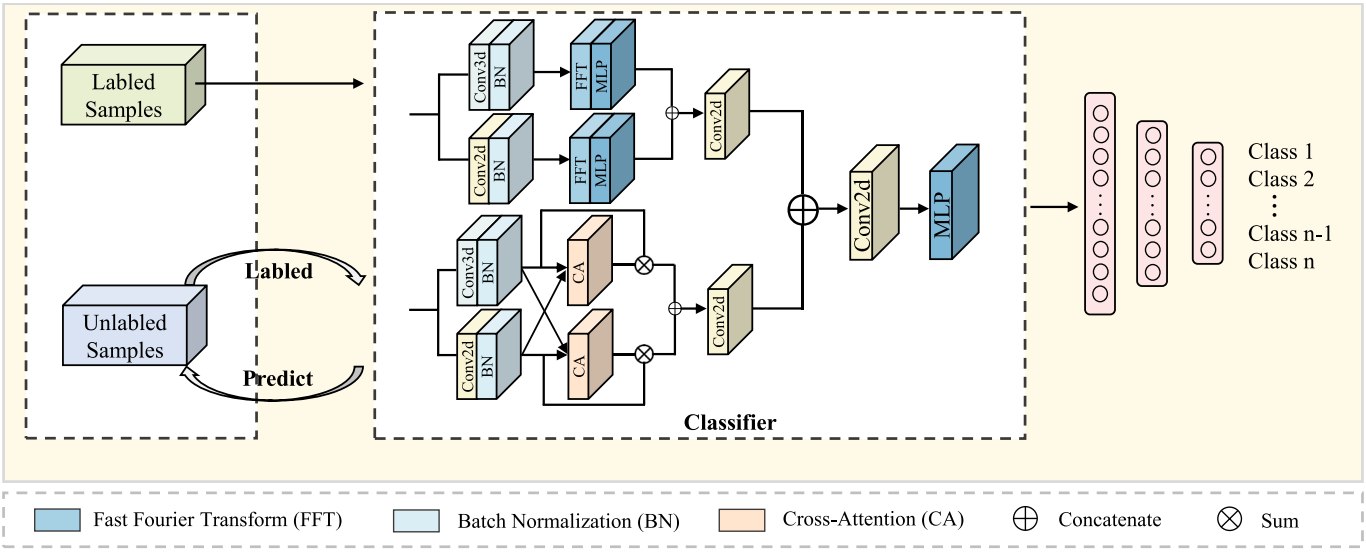


Fig. 4. Detailed workflow of STDNet. In the STDNet, which consists of the 3DCNN, the 2DCNN, the FFT, and the CAM. Our STDNet fully leverages the frequency, spectral, and spatial features of wheat spectral images by combining the frequency domain and CAMs and utilizing a cascade of 3-D and 2-D convolutional modules to improve classification accuracy. Specifically, 3DCNN is effective in jointly modeling spectral–spatial correlations across adjacent bands, while 2DCNN focuses on extracting refined spatial features from individual spectral slices.

Unlike existing methods that rely primarily on convolution or attention mechanisms, STDNet integrates spectral and temporal feature streams in a unified framework, enhancing feature discrimination, improving robustness across real, generated, and mixed datasets, and achieving consistently high classification accuracy. This makes STDNet a practical and effective solution for precise hyperspectral wheat variety identification.

III. Methodology

The workflow of our introduced hybrid supervised-driven two-stream collaborative network (HSTNet) for hyperspectral wheat variety classification is established in Fig. 2. It comprises an SCGAN (see Fig. 3) for data augmentation and a supervised two-stream discriminative network (STDNet in Fig. 4) for wheat classification. HSTNet integrates the adversarial collaboration of generation and classification through two collaborative tasks of generation and classification to achieve data augmentation and refined classification of wheat varieties. Specifically, the generation part is composed of a semi-supervised conditional adversarial generative network, while the classification part combines supervised and unsupervised modes to achieve hybrid supervised learning. In addition, Table I provides details of the network structure of this proposed HSTNet to help readers understand the structure of the whole network.

A. Data Preprocessing

The initial hyperspectral wheat images have a lot of noise and redundant features. Inspired by MOCO [51], we propose an adaptive weighted sliding window preprocessing method for hyperspectral image preprocessing. Specifically, we assume that the raw data sequence is $\mathbf{X} \in \mathbb{R}^{N\times D}$, where N is the number of samples, and D is the dimension of the sample features. We label the features of the jth sample as $\mathbf{X}_j \in \mathbb{R}^D$, and the window size as w. Subsequently, we define the smooth feature output matrix of each sample under the window size of w as

$$\tilde{\mathbf{X}}_j = \begin{cases} \dfrac{1}{w}\displaystyle\sum_{t=j}^{j+w-1} \mathbf{X}_t, & 1 < j \le N-w+1 \\ \mathbf{X}_j, & N-w+2 \le j \le N. \end{cases} \tag{1}$$

For the first $N-w+1$ samples, the output of the equation is the mean of the following w samples. Subsequently, the initial values are left unchanged for positions at the end that do not make a complete window. Meanwhile, we splice all smoothed samples to form the overall smoothing result matrix as

$$\tilde{\mathbf{X}} = \begin{bmatrix} \tilde{\mathbf{X}}_1 \\ \tilde{\mathbf{X}}_2 \\ \vdots \\ \tilde{\mathbf{X}}_N \end{bmatrix} \in \mathbb{R}^{N\times D}. \tag{2}$$

Although the sliding window averaging algorithm can partially improve data quality, it remains insufficient for raw hyperspectral data with high feature dispersion and severe noise. To address this, we propose an adaptive weighting mechanism as an optimization strategy to further denoise the smoothed hyperspectral image, defined as

$$\mathbf{W} = \frac{\sigma\left(\tilde{\mathbf{X}}\right)^2 + \sigma\left(\tilde{\mathbf{X}}\right)}{\max\left(\sigma\left(\tilde{\mathbf{X}}\right)^2\right) + \max\left(\sigma\left(\tilde{\mathbf{X}}\right)\right)} \tag{3}$$

where $\sigma(\tilde{\mathbf{X}})$ and $\sigma(\tilde{\mathbf{X}})^2$ denote the standard deviation and variance of $\tilde{\mathbf{X}}$, and $\mathbf{W}$ represents the calculated weights. Finally, we obtain the preprocessing feature $\mathbf{F}_{\text{pre}} = \tilde{\mathbf{X}} \odot \mathbf{W}$ by performing an element-by-element product of $\mathbf{W}$ and $\tilde{\mathbf{X}}$.

B. Semi-Supervised Conditional GAN

Leveraging the strengths of VAEs in learning fine-grained latent representations and GANs in generating high-quality,

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diverse samples, we propose an SCGAN. The generator, equipped with an LSTM module, receives random noise along with true labels and controls output fidelity through a label-constrained generation process. The generated samples are then input to a multilayer perceptron-based discriminator alongside real samples and their labels. An identification module evaluates the generated samples, verifying both their authenticity and category consistency to ensure high-quality outputs.

1) Generator With LSTM: The generator in Fig. 2(a) mainly consists of an encoder module and a decoder module. SCGAN utilizes LSTM not only to leverage memory cells for storing and transmitting long-term dependency information but also to address the issues of gradient vanishing or explosion commonly found in traditional GAN. We will introduce the design details of SCGAN below.

a) Encoder: During the initial training phase, the input noisy data conforming to a normal distribution can be represented as $\mathbf{X}_n \in \mathbb{R}^{L\times H\times W}$, where L denotes the number of sample bands, and $H \times W$ denotes the spatial dimension of the wheat hyperspectral image. The process of encoding $\mathbf{X}$ and wheat variety label $\mathbf{C}$ by LSTM in the encoding module is

$$\mathbf{F}_{\text{encode}} = \text{LSTM}\left(\text{repeat}\left(\text{cat}\left[\mathbf{X}_n, \mathbf{C}\right], 25\right)\right) \tag{4}$$

where cat[·] denotes the splicing of $\mathbf{X}_n$ and the labeling matrix $\mathbf{C}$ on a 1-D vector and repeating the input data 25 times to fill in to improve the completeness and stability of the feature information using repeat(·).

b) Decoder: Before the encoder output feature $\mathbf{F}_{\text{encode}}$ is fed into the decoder, it needs to be parameterized to improve the model's capacity to model uncertainty in the feature distribution. The following equation can represent the process as:

$$\sigma = \exp\left(\frac{\log\sigma^2\left(\mathbf{F}_{\text{encode}}\right)}{2}\right) \tag{5}$$

$$\mathbf{F}_{\text{rep}} = \mu\left(\mathbf{F}_{\text{encode}}\right) + \sigma \odot \varepsilon \tag{6}$$

where $\sigma^2(\cdot)$ and $\mu(\cdot)$ denote the variance and mean of $\mathbf{F}_{\text{encode}}$, respectively, and ε denotes sampling from a standard normal distribution to introduce randomness, which makes the random sampling process on the model tractable to facilitate gradient optimization in model training. The final obtained heavy parametric feature $\mathbf{F}_{\text{rep}}$ is used as an input to the decoder, and its decoding process as

$$\mathbf{F}_{\text{decode}} = \text{LSTM}\left(\text{repeat}\left(\text{cat}\left[\mathbf{F}_{\text{rep}}, \mathbf{C}\right], 25\right)\right). \tag{7}$$

2) Discriminator With Multilayer Perceptron: The discriminator effectively distinguishes between real samples and generated samples by learning the features of the samples. Meanwhile, its discriminant loss acts on the generator through backpropagation, prompting it to continuously optimize the generation strategy and learn a more robust feature distribution. Therefore, we input the data $\mathbf{F}_{\text{decode}}$ into the discriminator to determine whether it is true or false, and the output corresponding result matrix $\mathbf{F}_{\text{inp}}$ can be defined as

$$\mathbf{F}_{\text{inp}} = \sigma_{\text{out}}\left(W^{(L-1)}\left(\mathcal{O}_{k=K}^{1}\text{LR}\left(W^{(k)}\mathbf{F}_{\text{decode}}^{(k-1)} + b^{(k)}\right)\right) + b^{(L-1)}\right) \tag{8}$$

where K is the number of layers, σ_{out} denotes the output layer activation function Sigmoid(·), $W^{(k)}$ denotes a linear transformation of the input, $\mathcal{O}_{k=K}^{1}$ denotes a composite function operation starting at index K, LR(·) is a nonlinear activation LeakyReLU function, and $b^{(k)}$ is the bias term of the neuron.

C. Supervised Two-Stream Discriminative Network

To fully exploit the rich information in spectral data, we employ a hybrid training strategy that combines supervised and unsupervised learning. Supervised learning leverages labeled wheat samples to guide feature extraction, while unsupervised learning uncovers underlying patterns, correlations, and intrinsic structures within the spectral data.

1) Frequency Domain MultiLayer Perceptron: Fig. 2(b) depicts the structure of the frequency-domain fusion module. It consists of two sequential branch modules that independently convert the results of 3-D and 2-D convolution operations into the frequency domain. Using 3-D convolution as an example, the calculation formula is given by

$$\mathbf{F}_{i,j}^{l,h,w} = \delta_{i,j} + \sum_{c=1}^{C_n}\sum_{l=0}^{L}\sum_{h=0}^{H}\sum_{w=0}^{W} w_{i,j,c}^{l,h,w} \times \mathbf{F}_{\text{pre}} \tag{9}$$

where C is the input channel, c and l are the channel index and band index, respectively, h and w are the spatial dimension indexes of the convolution kernel, respectively, $\delta_{i,j}$ represents the bias term of the jth output feature map in the ith layer, and $w_{i,j,c}^{l,h,w}$ denotes the learnable convolution kernel weights of the feature map at the location (l, h, w). For the output feature $\mathbf{F}_{i,j}^{l,h,w}$, we utilize the ReLU(·) function for activation expressed as

$$\mathbf{F}_{l,h,w} = \text{Max}\left(0, \gamma_i \frac{\mathbf{F}_{i,j}^{l,h,w} - \mu_i}{\sqrt{\sigma_i^2 + \epsilon}} + \beta_i\right) \tag{10}$$

where γ_i and β_i are learnable scaling and translation parameters, and μ_i and σ_i^2 represent the mean and variance of all samples in the current data, respectively. In addition, ϵ is a minimal number introduced to prevent the denominator from being zero. To effectively mine the periodicity and global structure information in the image, the process of extracting the spectral feature $\mathbf{F}_{l,h,w}$ by applying the FFT to $\mathbf{F}_{\text{fft}}$ can be represented as

$$\mathbf{F}_{\text{fft}} = \text{FFT}\left(\mathbf{F}_{l,h,w}\right). \tag{11}$$

Specifically, the transform maps the feature $\mathbf{F}_{l,h,w}$ to the frequency domain, which helps suppress noise interference and highlight the texture structure of wheat. In addition, we normalized the $\mathbf{F}_{\text{fft}}$ to eliminate distribution differences among feature dimensions, thereby improving the training stability of the model. Subsequently, we adopt an MLP containing the GELU(·) function for nonlinear feature transformation to extract the expression of more discriminative high-level features as

$$\mathbf{F}_{\text{mlp}} = \text{MLP}\left(\text{LayerNorm}\left(\mathbf{F}_{\text{fft}}\right)\right). \tag{12}$$

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2) Cross-Attention Convolution (CAC): Fig. 2(b) gives the design related to CAC. The module separately extracts spectral and spatial features of wheat hyperspectral images via 3-D convolution, and texture and spatial features through 2-D convolution. Then, a cross-attention module integrates these 3-D and 2-D convolution outputs to refine the spectral, spatial, and texture features. Finally, the CAC is defined as

$$\mathbf{F}_{\text{att}} = \text{Softmax}\left(\frac{\mathbf{W}_Q\left(\text{Conv}_{2\text{d}}\left(\mathbf{F}_{\text{pre}}\right)\right)\left(\mathbf{W}_K\left(\text{Conv}_{3\text{d}}\left(\mathbf{F}_{\text{pre}}\right)\right)\right)^T}{\sqrt{d}}\right) \mathbf{W}_V\left(\text{Conv}_{3\text{d}}\left(\mathbf{F}_{\text{pre}}\right)\right) \tag{13}$$

where d represents the feature dimension, and T denotes the transpose operation of the implementation matrix. We utilize $\mathbf{F}_{\text{pre}}$ as a query (Q) in the attention mechanism after 2-D convolution and also as a key (K) and value (V) after 3-D convolution. It also maps $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$ to the uniform feature space by linear transformation. Finally, we perform the cross-attention computation and regression operations using the Softmax($\cdot$) function.

D. Loss Function

In this study, we designed dedicated loss functions for each module to balance generation and classification, ensuring stable training of the generator, discriminator, and classifier. The generator loss combines reconstruction, Kullback–Leibler (KL) divergence, adversarial, and classifier-guided losses to prevent gradient explosion and mode collapse common in GANs. The discriminator loss accounts for both real and generated samples, while the classifier loss is based on cross-entropy. The overall HSTNet objective integrates the generation, discrimination, and classification losses into a unified optimization framework.

1) Generator Loss: The generator loss quantifies how convincingly the discriminator classifies generated samples as real, with lower values indicating better generator performance. To enhance reconstruction quality, we adopt an explicit reconstruction mechanism inspired by VAEs, which improves the detail fidelity of generated samples and compensates for the limitations of purely adversarial loss in capturing fine-grained features. Specifically, the reconstruction loss, defined as the mean squared error, is expressed as

$$\text{Loss}_{\text{mse}} = \mathbb{E}_{x\sim p(x)}\left[\|x - \hat{x}\|^2\right] \tag{14}$$

where x is the true sample, $\hat{x}$ is the generated sample, and $\mathbb{E}$ is the expectation. Subsequently, we incorporated the KL divergence loss to emphasize the difference between the latent variable distribution and the standard normal distribution. The KL divergence loss is defined as

$$\text{Loss}_{\text{kl}} = \sum_{i=1}^{N} p\left(x_i\right)\log\left(\frac{p\left(x_i\right)}{q\left(x_i\right)}\right) \tag{15}$$

where $p(x_i)$ and $q(x_i)$ represent the true probability distribution and the predicted probability distribution, respectively. The preceding process addresses the self-reconstruction component of the generator. Drawing on adversarial generation principles, we incorporate the discriminators' adversarial loss to guide the generator, enabling it to learn latent features more effectively and produce more realistic samples. Mathematically, the adversarial loss is defined as

$$\begin{aligned}\text{Loss}_{\text{adv}} = {} & \mathbb{E}_{x\sim P(x)}\left[\log D\left(x\right)\right] \\ & + \mathbb{E}_{z\sim P(z)}\left[\log\left(1 - D\left(G\left(z\right)\right)\right)\right]\end{aligned} \tag{16}$$

where x is the sample from the real data distribution $P(x)$; z is the input from the noise distribution $P(z)$; $G(z)$ is the false sample generated by the generator; $D(x)$ and $D(G(z))$ are the output probabilities of the discriminator for the real sample and the false sample; and $\mathbb{E}$ represents the expected operation. In addition, we added the auxiliary generation loss of the classifier in our experiments to allow the generator to produce higher quality data, and the loss is defined as

$$\text{Loss}_{\text{da}} = -\sum_{i=1} y_i \log\left(C\left(G\left(\mathbf{z}\right)\right)_i\right) \tag{17}$$

where y_i is the fixed target label generated by the generator; and $C(G(\mathbf{z}))_i$ is the predicted probability of the classifier for the ith category of the generator output $G(\mathbf{z})$. In summary, the final loss function of the generator can be expressed as

$$\text{Loss}_G = \text{Loss}_{\text{mse}} + \text{Loss}_{\text{kl}} + \text{Loss}_{\text{adv}} + \text{Loss}_{\text{da}}. \tag{18}$$

2) Discriminator Loss: The discriminator performs binary classification by distinguishing real from generated samples, measuring the divergence between the true and generated data distributions. Accordingly, separate losses are computed for real and generated data. The loss for real samples as

$$\text{Loss}_D^{\text{real}} = -\mathbb{E}_{(x,y)\sim P(x)}\left[\log D\left(x, y\right)\right] \tag{19}$$

where y denotes the true data label. For generated data, we want the discriminator to identify the generated data as false as possible. Therefore, the loss function minimizes the output probability as much as possible to guide the generator to evolve toward the target. The loss for generated data is calculated as

$$\text{Loss}_D^{\text{fake}} = -\mathbb{E}_{\left(x_g, y_g\right)\sim P_g}\left[\log\left(1 - D\left(x_g, y_g\right)\right)\right] \tag{20}$$

where P_g is the generated data distribution and x_g and y_g denote the data and labels generated by generator G. The final composition discriminator loss is defined as

$$\text{Loss}_D = \text{Loss}_D^{\text{real}} + \text{Loss}_D^{\text{fake}}. \tag{21}$$

3) Classifier Loss: The classifier captures latent features by minimizing the discrepancy between predicted probabilities and true labels, enabling accurate classification of the 20 wheat categories. To quantify this difference and maximize classification accuracy, we employ the cross-entropy loss, defined as

$$\text{Loss}_C = -\sum_{i=1}^{N}\sum_{j=1}^{M} y_{ij}\log\left(\hat{y}_{i,j}\right) \tag{22}$$

where M is the number of categories, $y_{i,j}$ is the true label of the ith sample, and $\hat{y}_{i,j}$ is the predicted probability of the jth category for the ith sample.

Eventually, the overall objective loss function of HSTNet is formulated as

$$\text{Loss} = \text{Loss}_G + \text{Loss}_D + \text{Loss}_C. \tag{23}$$

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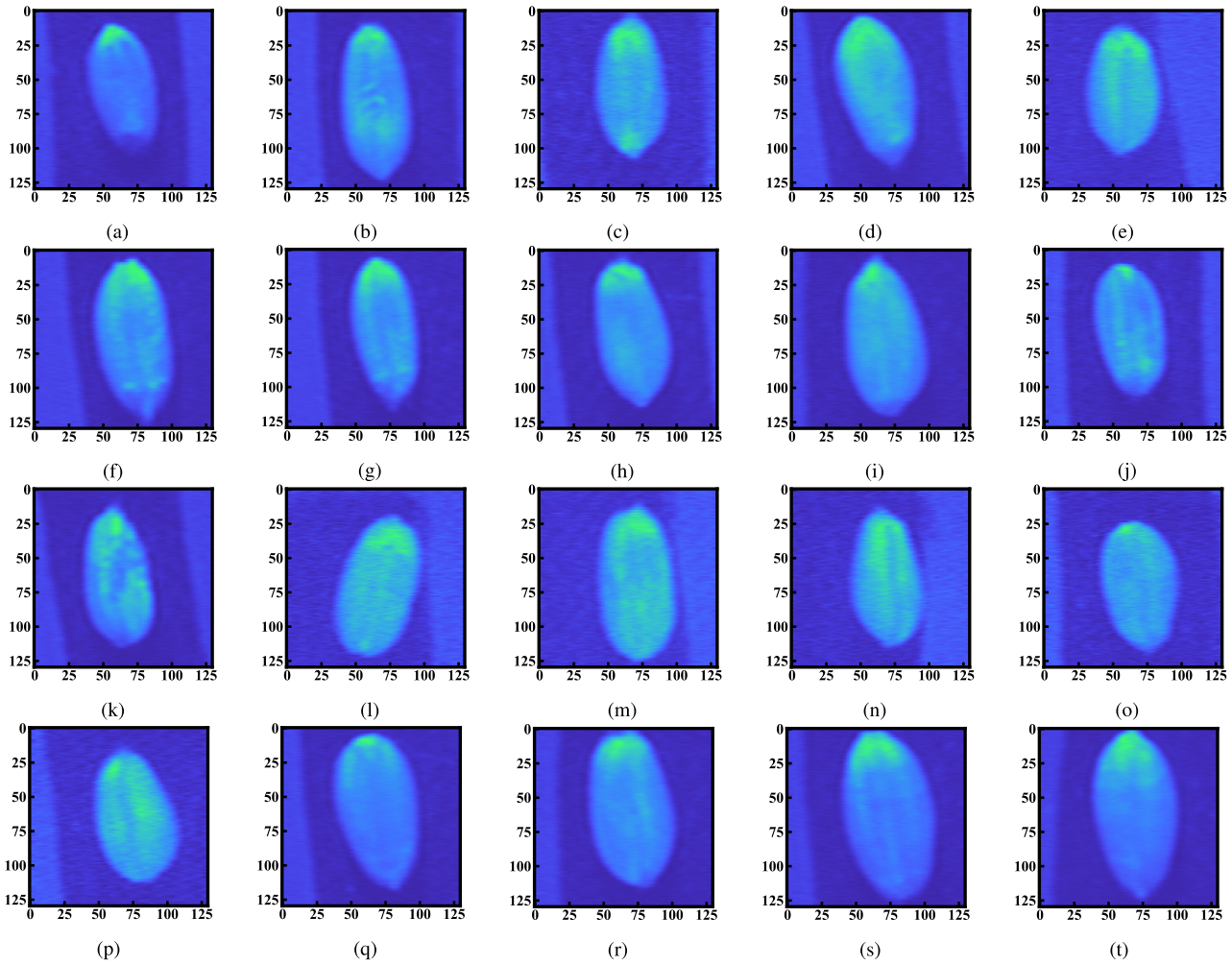


Fig. 5. Spectral images of 20 varieties of wheat seeds in the 80th band. (a) 80th band spectrogram of Jimai 22. (b) 80th band spectrogram of Zhengmai 101. (c) 80th band spectrogram of Yuyuan 916. (d) 80th band spectrogram of Zhongmai 578. (e) 80th band spectrogram of Zhoumai 28. (f) 80th band spectrogram of Fengdecun 21. (g) 80th band spectrogram of Bainong 4199. (h) 80th band spectrogram of Xinong 511. (i) 80th band spectrogram of Bainong 307. (j) 80th band spectrogram of Zhengmai 366. (k) 80th band spectrogram of Zhoumai 36. (l) 80th band spectrogram of Zhongyu 9307. (m) 80th band spectrogram of Luomai 28. (n) 80th band spectrogram of Zhoumai 27. (o) 80th band spectrogram of Shangmai 167. (p) 80th band spectrogram of Xinmai 32. (q) 80th band spectrogram of Bainong 207. (r) 80th band spectra of Banong AK 58. (s) 80th band spectra of Zhengmai 7698. (t) 80th band spectra of Zhongmai 18.

IV. Experiment and Analysis

In this section, we begin by detailing the dataset creation and experimental configuration. Next, we compare and analyze the performance of our HSTNet against ten leading methods on the constructed MWHD dataset. Finally, we perform an ablation study to assess the impact of each module on HSTNets' classification accuracy.

A. Mixed Wheat Hyperspectral Dataset

We created an MWHD containing 20 wheat varieties with a total of 4800 samples, which contains 2400 raw wheat spectral samples and 2400 wheat spectral samples generated utilizing our HSTNet.

For 2400 wheat hyperspectral images, we used a SOC 710 portable visibleCireCinfrared spectrometer to capture spectral images of both the front and back of the seeds. The spectrometer covers a range of 400–1000 nm with a spectral resolution of 4.6875 nm across 128 bands. The images were captured at a resolution of 696 $\times$ 520 pixels and then cropped to isolate the seed area, yielding individual samples with a spatial resolution of 80 $\times$ 110 pixels. In addition, Fig. 5 reports images of the 80th band in hyperspectral imaging corresponding to 20 classes of wheat seeds.

For 2400 generated wheat hyperspectral images, it is produced by SCGAN, learning the data and spatial distribution of the original wheat hyperspectral samples. Concretely, SCGAN first trains the generator in the network in a semi-supervised manner, which feeds the generator with noisy data that obeys a normal distribution and combines real data labels as constraints. The discriminator discriminates between the generator's output and the real data to construct a complete conditional adversarial generative network. As shown in Fig. 6, the generated data exhibits a high degree of consistency with the raw data.

B. Experimental Configuration

HSTNet is extensively trained and tested on our constructed MWHD. Concretely, HSTNet was tested on 2400 real wheat

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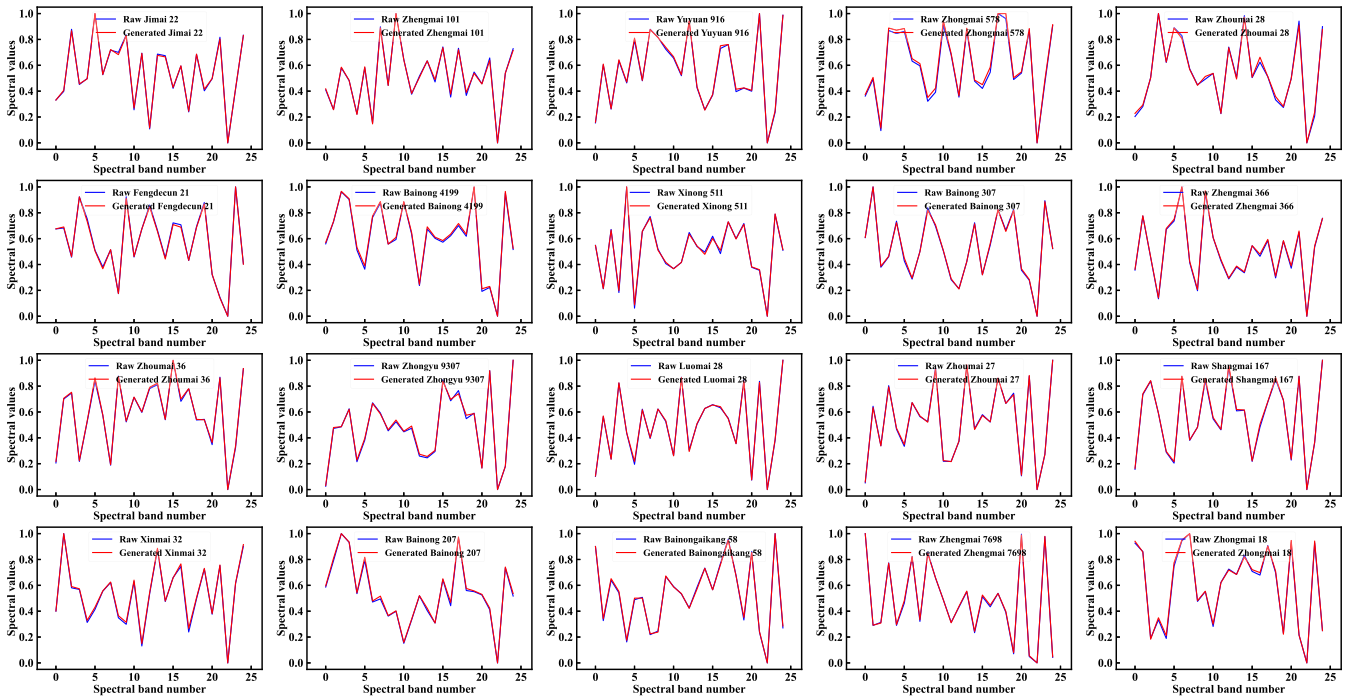


Fig. 6. Mean folds of the spectral data samples generated by SCGAN and the real spectral samples of the corresponding species in the 0–25 band. It can be seen that the generated samples and the real sample's mean folds almost coincide, which fully reflects the better generation effect of SCGAN.

samples, 2400 generated wheat samples, and 4800 samples of real wheat varieties mixed with generated wheat varieties. In each dataset, our training samples and test samples are configured in a 4:1 ratio. The HSTNet is performed on an Ubuntu 24.04 LTS with an Intel Core i7-13700KF 16-Core CPU running at 3.4 GHz, 32-GB memory, an NVIDIA GeForce RTX 4070 Ti Super GPU, and the PyTorch Deep Learning Toolbox. HSTNet consists of an SCGAN and an STDNet. For SCGAN training, a batch size is set to 64, the optimizer uses Adam with decay parameters set to 0.6 and 0.999, a learning rate of 1×10^{-3}, and a total of 5000 training iterations. For STDNet training, a batch size is set to 64, the optimizer uses AdamW with decay parameters set to 0.5 and 0.999, a learning rate of 3×10^{-3}, and 5000 training iterations.

C. Evaluation Metrics

To comprehensively evaluate the performance of HSTNet on wheat hyperspectral image data, we employ the accuracy rate (A_T), recall rate (A_R), precision (A_P), $F1$-score (F_1), OA, AA, and Kappa as the primary evaluation metrics. In practical classification work, the classification results are classified as predicted positive and true positive (TP), predicted negative but TP (false negative, FN), predicted positive but true negative [TN, false positive (FP)], and predicted negative and TN. Specifically, A_R measures the model's capacity to identify samples of a specific category, A_P reflects the reliability of the prediction results, F_1 integrates the balance between A_R and A_P, and A_T indicates the overall classification accuracy of the classification method. The OA is used to measure the proportion of correctly classified samples out of the total sample size. The AA is calculated by determining the classification accuracy for each category and then taking the average. To further measure the consistency of model performance across categories, we introduce the Kappa coefficient as a supplementary assessment metric to evaluate the robustness of each classification method.

D. Classification Results

To thoroughly evaluate the classification performance of HSTNet across different wheat varieties, we conducted extensive experiments and comparative analyses against 12 classical deep learning methods on the real, generative, and mixed datasets. Specifically, these 12 methods include CNN-based methods (DSNet [52], 3DCNN [53], SDEnet [54], RSSAN [50], DFFN [55], S^{2}ENet [11], and SSEFN [56]), transformer-based methods (SSFTT [4] and MFT [57]), RNN-based method (BLSTM [25]), and attention-based methods (SSTNet [58] and PCANet [59]). To ensure fairness and objectivity in the experimental evaluation, we used the source code and parameter settings provided by the original authors for all compared methods.

1) Classification Evaluation of the Real Dataset: Tables II and III report the quantitative metric scores for our HSTNet experiments with 12 deep learning methods on real, generated, and hybrid datasets. For traditional CNN methods, 3DCNN [53], RSSAN [50], and DFFN [55] have poor classification performance. The fundamental reason is that their limited modeling capabilities make it difficult to efficiently learn and integrate the complex spectral and spatial dependencies in spectral data. However, the latest deep learning method SSEFN [56] employs a spatial–spectral enhancement strategy to enhance feature discrimination from both spatial and spectral perspectives. It utilizes an adaptive decision fusion module

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TABLE II

CLASSIFICATION RESULT SCORES FOR OUR HSTNET WITH 12 DEEP LEARNING METHODS EXPERIMENTED ON THREE WHEAT HYPERSPECTRAL DATASETS (NOTABLY, THE HIGHEST SCORES ARE DENOTED IN RED, AND THE NEXT HIGHEST SCORES ARE DENOTED IN BLUE)

Methods	Real dataset				Generated dataset				Mixed dataset			
	A_P	A_R	A_T	F_1	A_P	A_R	A_T	F_1	A_P	A_R	A_T	F_1
DSNet [52]	98.04±0.12	97.91±0.15	97.92±0.13	97.92±0.17	97.18±0.18	97.03±0.15	97.04±0.2	97.03±0.26	99.21±0.05	99.17±0.12	99.16±0.08	99.17±0.11
3DCNN [53]	95.83±0.27	95.72±0.28	95.71±0.32	95.71±0.19	94.33±0.35	94.17±0.37	94.18±0.35	94.18±0.28	96.54±0.47	96.33±0.31	96.34±0.34	96.33±0.11
SDEnet [54]	97.61±0.15	97.54±0.27	97.53±0.35	97.54±0.1	96.33±0.24	96.14±0.77	96.13±0.25	96.14±0.19	99.03±0.39	98.96±0.17	98.95±0.12	98.96±0.39
RSSAN [50]	96.43±0.28	96.33±0.21	96.33±0.15	96.34±0.38	95.89±0.14	95.64±0.31	95.65±0.33	95.65±0.09	98.01±0.14	97.91±0.18	97.91±0.58	97.92±0.25
DFFN [55]	96.06±0.29	96.54±0.36	96.62±0.17	96.54±0.29	95.33±1.68	95.18±1.82	95.18±1.75	95.19±0.69	97.79±0.98	97.48±1.15	97.49±1.07	97.48±1.12
S^2ENet [11]	97.21±1.17	97.08±0.35	97.09±1.28	97.09±1.31	97.01±0.79	96.91±1.63	96.92±1.57	96.91±0.61	98.33±0.56	98.08±0.72	98.09±0.38	98.09±0.75
SSEFN [56]	98.45±0.12	98.42±0.07	98.42±0.15	98.41±0.09	96.94±0.38	96.76±0.51	96.75±0.45	96.77±0.39	99.28±0.09	99.22±0.17	99.21±0.13	99.21±0.08
SSFTT [4]	97.25±0.23	97.12±0.41	97.13±0.37	97.13±0.45	96.25±1.19	96.03±1.02	96.03±1.05	96.04±2.01	98.17±0.69	97.92±0.78	97.91±0.83	97.92±0.76
MFT [57]	97.13±1.15	96.92±0.61	96.91±0.57	96.93±1.03	96.88±1.12	96.73±0.85	96.72±0.79	96.73±0.81	98.16±0.24	97.91±0.15	97.91±0.79	97.92±0.12
BLSTM [25]	98.06±0.23	97.88±0.37	97.89±0.19	97.88±0.62	97.96±0.63	97.81±0.52	97.82±0.87	97.81±0.31	98.64±0.15	98.54±0.12	98.55±0.28	98.54±0.05
SSTNet [58]	97.81±0.79	97.95±0.28	97.96±0.32	97.96±0.19	97.12±0.53	96.89±1.67	96.88±0.71	96.89±0.65	98.27±0.13	98.12±0.71	98.14±0.17	98.12±0.34
PPCANet [59]	97.98±0.22	97.92±0.25	97.92±0.38	97.92±0.11	97.48±0.27	97.08±0.43	97.12±0.79	97.08±0.45	99.13±0.06	99.04±0.13	99.04±0.19	99.03±0.05
HSTNet	99.03±0.07	98.96±0.12	98.97±0.09	98.96±0.13	98.28±0.12	98.12±0.16	98.14±0.17	98.12±0.11	99.63±0.05	99.54±0.02	99.53±0.09	99.54±0.07

TABLE III

CLASSIFICATION PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THREE DATASETS

Methods	Real dataset			Generated dataset			Mixed dataset		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
DSNet [52]	97.20±1.56	96.91±1.83	97.82±1.09	97.04±1.72	97.03±1.65	96.95±1.88	99.16±0.52	99.17±0.48	99.12±0.63
3DCNN [53]	95.71±0.92	94.72±1.15	95.58±0.87	94.18±1.36	94.17±1.42	94.09±1.53	96.34±0.75	95.53±0.92	96.06±0.88
SDEnet [54]	97.53±0.89	96.53±1.24	97.46±0.97	96.13±1.42	96.14±1.35	96.01±1.52	98.95±0.63	97.96±1.08	98.89±0.71
RSSAN [50]	96.63±1.05	95.54±1.32	96.24±1.18	95.65±1.27	95.64±1.31	95.58±1.34	97.91±1.02	97.91±1.04	97.83±1.09
DFFN [55]	96.62±1.07	95.58±1.33	96.19±1.21	95.12±1.43	95.18±1.39	95.04±1.47	97.89±1.05	96.88±1.57	97.35±1.26
S^2ENet [11]	97.09±1.18	96.04±1.52	96.93±1.14	95.98±1.51	95.91±1.54	96.73±1.21	99.09±0.62	99.04±0.64	97.93±1.02
SSEFN [56]	98.42±0.75	97.42±1.03	98.12±0.84	96.75±1.16	96.76±1.13	96.39±1.35	99.21±0.51	99.22±0.48	99.15±0.57
SSFTT [4]	97.53±0.91	96.12±1.48	97.02±1.13	96.03±1.48	96.03±1.50	95.85±1.57	97.50±1.15	95.91±1.87	97.76±1.03
MFT [57]	97.91±0.82	95.92±1.56	96.79±1.23	96.73±1.15	96.13±1.45	96.58±1.24	97.11±1.32	95.69±1.94	97.81±1.05
BLSTM [25]	97.10±1.15	96.20±1.47	97.79±1.05	97.82±1.04	97.81±1.02	97.67±1.08	98.54±0.93	97.54±1.18	98.46±0.91
SSTNet [58]	96.61±1.03	95.82±1.63	97.82±1.07	96.88±1.11	96.98±1.05	96.81±1.12	98.14±1.23	98.12±1.25	98.02±1.26
PPCANet [59]	97.12±1.11	97.90±0.97	97.69±1.04	97.12±1.10	97.09±1.14	97.67±1.08	99.00±0.72	99.04±0.65	98.92±0.81
HSTNet	98.97±0.62	97.96±0.89	98.91±0.74	98.14±0.89	98.12±0.93	98.02±0.96	99.53±0.32	99.54±0.31	99.41±0.42

to integrate all enhanced features, thereby achieving superior classification performance. For transformer-based methods, SSFTT [4] and MFT [57] both demonstrate good classification performance, which is superior to that of RSSAN [50] and DFFN [55]. For attention methods, SSTNet [58] proposes a spatial, spectral, and texture-aware attention network that further refines spatial and texture features by embedding an attention mechanism, yielding good results. Although SSEFN [56], BLSTM [25], and PCANet [59] have achieved good results for wheat variety classification, they are still inferior to our proposed HSTNet. In contrast, the HSTNet we proposed not only takes into account spectral and spatial features but also spectral information. It outperforms existing methods in multiple evaluation indicators, verifying its advancement and effectiveness in hyperspectral image classification tasks.

2) Classification Evaluation of the Generated Dataset: For the generated dataset, the classification scores of each comparison method show a decreasing trend to varying degrees compared with the real dataset. This performance degradation is primarily attributed to the fact that the feature selection mechanism of the model in the adversarial generation process inevitably overlooks some subtle features, resulting in a certain degree of loss in feature diversity of the generated data compared to the original data. 3DCNN [53], SDEnet [54], RSSAN [50], DFFN [55], SSEFN [56], SSFTT [4], and MFT [57] did not yield satisfactory performance in the classification of wheat hyperspectral images. Subsequently, DSNet [52], S^2ENet [11], SSTNet [58], and PPCANet [59] showed some improvement in their component performance compared to each other, yet remained inferior to our HSTNet. Among the RNN-based methods, BLSTM [25] fully explores the bidirectional spectral correlation in HSI, emphasizes the spatial spectral context within pixels, and reduces redundant information, thereby significantly improving classification performance and achieving good results. Although the recognition accuracy has decreased, our HSTNet still obtains the best classification results in various indicators.

3) Classification Evaluation of the Mixed Dataset: Compared to real or generated datasets, hybrid datasets exhibit significant improvements in all performance metrics. Among the CNN-based methods, SSEFN [56] still achieves excellent classification performance due to its unique feature enhancement strategy, and DSNet [52] significantly improves classification performance by fusing subpixel information and convolutional category features. 3DCNN [53] consistently performs poorly due to its insufficient consideration

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TABLE IV
EACH METHOD WAS TRAINED ON THE REAL DATASET AND CROSS-VALIDATED ON THE GENERATED AND MIXED DATASETS FOR CLASSIFICATION RESULTS

Methods	Real dataset				Generated dataset				Mixed dataset			
	A_P	A_R	A_T	F_1	A_P	A_R	A_T	F_1	A_P	A_R	A_T	F_1
DSNet [52]	98.04±0.12	97.91±0.15	97.92±0.13	97.92±0.17	93.81±0.12	92.55±0.11	92.93±0.14	93.41±0.18	98.43±0.04	98.21±0.10	98.45±0.07	98.33±0.09
3DCNN [53]	95.83±0.27	95.72±0.28	95.71±0.32	95.71±0.19	90.55±0.24	89.91±0.25	90.41±0.28	90.77±0.19	95.88±0.41	95.35±0.29	95.61±0.25	95.44±0.08
SDEnet [54]	97.61±0.15	97.54±0.27	97.53±0.35	97.54±0.1	92.64±0.16	91.88±0.51	91.59±0.21	92.11±0.14	98.11±0.31	98.17±0.15	98.02±0.10	98.01±0.27
RSSAN [50]	96.43±0.28	96.33±0.21	96.33±0.15	96.34±0.38	91.43±0.09	91.11±0.23	91.95±0.19	91.59±0.07	97.28±0.12	96.95±0.16	97.13±0.49	97.11±0.21
DFFN [55]	96.06±0.29	96.54±0.36	96.62±0.17	96.54±0.29	91.82±1.13	90.71±1.25	90.81±1.44	91.75±0.45	96.93±0.81	96.55±0.92	96.33±0.81	96.88±0.91
S^2ENet [11]	97.21±1.17	97.08±0.35	97.09±1.28	97.09±1.31	92.68±0.44	93.77±1.31	92.83±0.99	92.19±0.39	97.61±0.41	97.31±0.58	97.13±0.31	97.22±0.61
SSEFN [56]	98.45±0.12	98.42±0.07	98.42±0.15	98.41±0.09	92.21±0.21	92.91±0.39	92.01±0.33	92.64±0.25	98.55±0.07	98.31±0.14	98.37±0.11	98.41±0.06
SSFTT [4]	97.25±0.23	97.12±0.41	97.13±0.37	97.13±0.45	91.89±0.73	91.66±0.81	91.52±0.77	92.11±1.31	97.11±0.61	96.88±0.65	96.99±0.58	97.03±0.55
MFT [57]	97.13±1.15	96.92±0.61	96.91±0.57	96.93±1.03	92.41±0.93	92.17±0.55	91.99±0.51	92.68±0.59	97.35±0.20	97.11±0.13	97.05±0.52	97.19±0.61
BLSTM [25]	98.06±0.23	97.88±0.37	97.89±0.19	97.88±0.62	93.11±0.39	93.22±0.33	93.61±0.52	93.14±0.23	97.88±0.13	97.64±0.11	97.71±0.22	97.69±0.04
SSTNet [58]	97.81±0.79	97.95±0.28	97.96±0.32	97.96±0.19	92.33±0.38	92.71±0.99	92.01±0.49	92.28±0.49	97.33±0.11	97.29±0.59	97.31±0.11	97.41±0.08
PPCANet [59]	97.98±0.22	97.92±0.25	97.92±0.38	97.92±0.11	93.01±0.18	92.77±0.31	92.55±0.58	92.61±0.38	98.15±0.05	98.31±0.11	98.11±0.16	98.22±0.04
HSTNet	99.03±0.07	98.96±0.12	98.97±0.09	98.96±0.13	93.81±0.07	94.21±0.12	94.33±0.12	93.75±0.07	98.71±0.04	98.66±0.02	98.61±0.07	98.59±0.06

TABLE V
CLASSIFICATION PERFORMANCE COMPARISON ON UNIVERSITY OF PAVIA DATASET

Methods	A_P	A_R	A_T	F_1	OA	AA	Kappa
DSNet [52]	97.70±1.23	97.65±1.35	97.65±1.28	97.61±1.31	98.74±0.52	97.33±1.42	97.38±1.37
3DCNN [53]	74.99±3.51	82.76±2.83	78.82±3.17	78.93±3.24	77.14±3.45	80.08±2.96	77.32±3.38
SDEnet [54]	95.70±1.85	94.94±1.92	94.95±1.87	95.07±1.91	93.46±2.05	95.52±1.83	94.78±1.94
RSSAN [50]	98.55±0.82	99.19±0.63	98.87±0.75	98.87±0.72	97.01±1.13	98.63±0.68	98.79±0.65
DFFN [55]	97.57±1.27	97.44±1.32	97.33±1.35	97.30±1.38	96.07±1.42	97.24±1.31	97.15±1.36
S^2ENet [11]	97.75±1.21	97.62±1.25	97.62±1.28	97.31±1.32	98.22±0.68	96.79±1.35	97.22±1.31
SSEFN [56]	98.81±0.75	98.82±0.72	98.82±0.74	98.79±0.71	97.66±1.02	96.64±1.42	98.33±0.75
SSFTT [4]	95.19±1.93	94.71±1.98	94.71±1.95	94.44±2.01	96.58±1.37	96.19±1.45	94.18±1.97
MFT [57]	98.79±0.85	98.78±0.87	98.78±0.86	98.77±0.88	97.77±1.05	98.48±0.75	98.55±0.82
BLSTM [25]	92.99±3.21	92.85±3.27	92.85±3.24	92.89±3.26	94.65±3.13	93.73±3.25	92.76±3.31
SSTNet [58]	96.92±1.53	96.89±1.55	96.91±1.54	96.92±1.56	98.38±0.72	95.91±1.63	96.22±1.58
PPCANet [59]	95.10±1.95	95.14±1.93	95.15±1.94	94.83±2.02	93.44±2.07	94.06±1.98	94.45±1.96
HSTNet	98.84±0.68	98.82±0.69	98.91±0.65	98.92±0.67	98.81±0.55	97.95±0.82	98.81±0.63

of spatial data in hyperspectral images. In the transformer-based method, MFT [57] achieves better performance by designing a multimodal fusion transformer to learn unique representations in a compact and hierarchical feature space. In contrast, our HSTNet achieves multidimensional feature fusion by integrating spatial, spectral, and signal spectrum features, which significantly improves the generalization and feature recognition capabilities of the model and enables it to perform well in mixed dataset classification tasks. Even marginal improvements beyond 99% accuracy are meaningful in large-scale breeding and agricultural production, where a small percentage of misclassification could translate into significant losses in yield estimation and resource allocation.

4) Cross-Validated on the Generated and Mixed Datasets: Our HSTNet experiments trained multiple deep learning models on real datasets, with cross-validation conducted separately on real, generated, and mixed datasets. In Table IV, HSTNet demonstrated superior performance across all evaluation scenarios, thereby validating its exceptional feature extraction and generalization capabilities. HSTNet outperformed other methods by over 98.9% across all metrics on the real dataset. SSEFN [56] maintained metrics above 98% on both the real and mixed datasets, achieving the second-best overall results. It should be noted that the performance of each method generally declined by 2%–6% on the generated dataset, reflecting the distributional differences between the real and generated domains. Despite this performance drop, HSTNet maintained optimal metrics across all metrics, demonstrating its superior domain adaptation capability. It is worth noting that some models, such as BLSTM [25] and S^2ENet [11], show relative advantages in generating datasets. Experimental results on mixed datasets validate the effectiveness of data augmentation strategies. HSTNet achieves an F_1 of 98.59% on the mixed dataset, which is close to its performance on the real dataset and demonstrates that judicious data fusion effectively mitigates domain adaptation challenges. In addition, the performance of most models on mixed datasets falls between that on real and generated datasets, consistent with the anticipated performance interpolation effect. Experimental results demonstrate that HSTNet's network architecture design exhibits significant advantages and generalization capabilities when processing multidomain data.

5) Classification Evaluation of the publicly Available of the University of Pavia Dataset: To demonstrate the generalization performance of our HSTNet method, we experimentally evaluated HSTNet on the publicly available University of Pavia dataset with 12 comparison methods. The dataset was acquired by the ROSIS sensor located in Pavia, Italy, and contains 103 spectral bands in the wavelengths range from 0.43 to 0.86 μm with a spatial resolution of 610 $\times$ 340 pixels.

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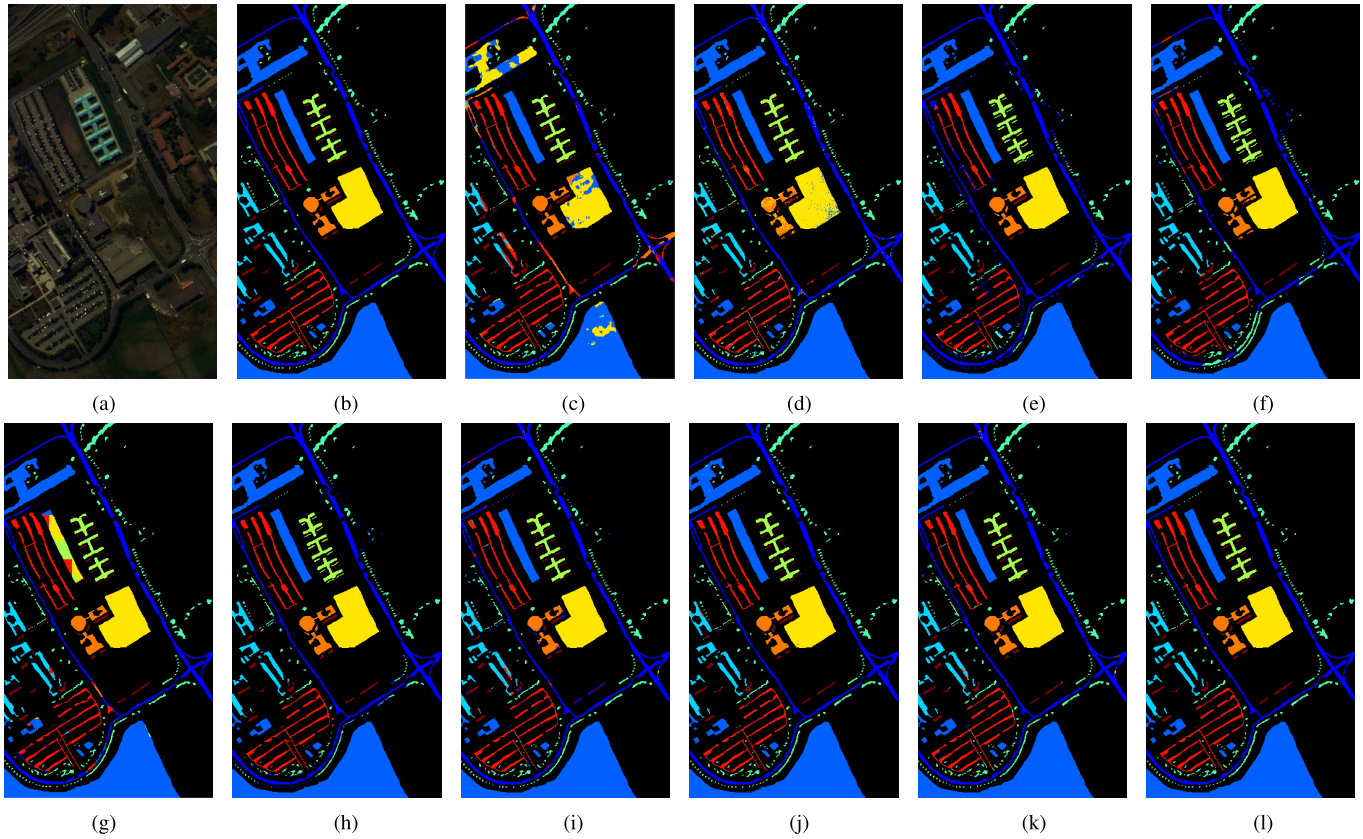


Fig. 7. Classification maps of different methods on the University of Pavia dataset. (a) False-color image. (b) Ground truth. (c) 3DCNN [53]. (d) BLSTM [25]. (e) SSFTT [4]. (f) PPCANet [59]. (g) S2ENet [11]. (h) DFFN [55]. (i) DSNet [52]. (j) RSSAN [50]. (k) SSEFN [56]. (l) HSTNet.

TABLE VI

ABLATION STUDY OF KEY MODULES IN HSTNET (✓ = MODULE RETAINED, BLANK = MODULE ABLATED)

Cases	DPP	SSCGAN	FDMP	CACM	A_P	A_R	A_T	F_1	K_A
1		✓	✓	✓	93.49±2.21	93.33±3.76	93.33±3.05	93.34±4.86	93.31±3.89
2	✓		✓	✓	94.25±4.11	94.12±2.68	94.13±3.67	94.12±3.98	93.96±3.04
3	✓	✓		✓	98.24±1.01	98.13±0.37	98.12±0.08	98.13±1.22	97.99±0.20
4	✓	✓	✓		98.81±1.09	98.85±1.01	98.75±0.11	98.74±1.14	98.63±0.26
5 (13)	✓	✓	✓	✓	94.71±1.55	94.57±1.30	94.56±0.15	94.57±1.03	94.39±1.37
5 (15)	✓	✓	✓	✓	96.76±1.35	96.48±1.10	96.44±1.21	96.42±1.52	96.32±1.27
5 (19)	✓	✓	✓	✓	97.32±0.21	97.17±0.33	97.17±0.70	97.20±0.91	97.18±1.06
5 (1-MLP)	✓	✓	✓	✓	98.58±0.10	98.39±0.25	98.38±0.18	98.39±0.07	98.10±0.21
5 (3-MLP)	✓	✓	✓	✓	98.84±0.30	98.63±0.22	98.62±0.10	98.63±0.11	98.31±0.32
5 (4-MLP)	✓	✓	✓	✓	98.61±0.08	98.55±0.15	98.52±0.11	98.55±0.37	98.22±0.10
5 (2DCNN)	✓	✓	✓	✓	95.35±0.63	95.33±0.79	95.11±0.31	95.1±0.15	94.89±0.58
5 (3DCNN)	✓	✓	✓	✓	97.58±0.37	97.59±0.30	97.31±0.27	97.32±0.20	97.18±0.29
5 (17, 2, 2DCNN, 3DCNN)	✓	✓	✓	✓	99.03±0.07	98.96±0.12	98.97±0.09	98.96±0.13	98.91±0.31

The dataset includes nine different land cover classes. For this dataset, both the training and test data were partitioned in the ratio of 7 : 3. From Table V, we can observe that in our HSTNet experiments on publicly available remote sensing datasets, our HSTNet achieves the highest or near-highest scores for A_T, A_R, A_P, F_1, OA, AA, and Kappa in most cases of remote sensing image classification. Meanwhile, we give the visual classification results of HSTNet and various methods evaluated on the public University of Pavia dataset in Fig. 7. From the visual classification Fig. 7, SSEFN [56] and RSSAN [50] achieve excellent classification performance by enhancing feature discrimination from both spatial and spectral perspectives. On the other hand, 3DCNN [53]'s method has a poor performance in fusing spatial features with spectral information, resulting in some linear feature breaks and boundary inconsistencies. In contrast, our HSTNet achieves multidimensional feature fusion by integrating spatial features, spectral features, and signal spectrum features, and achieves better classification performance than other methods. Overall, our HSTNet has good classification performance and generalization performance on both our constructed MWHD and publicly available remote sensing datasets.

In addition, Fig. 8 illustrates the relationship between the training loss of HSTNet on various datasets and the accuracy

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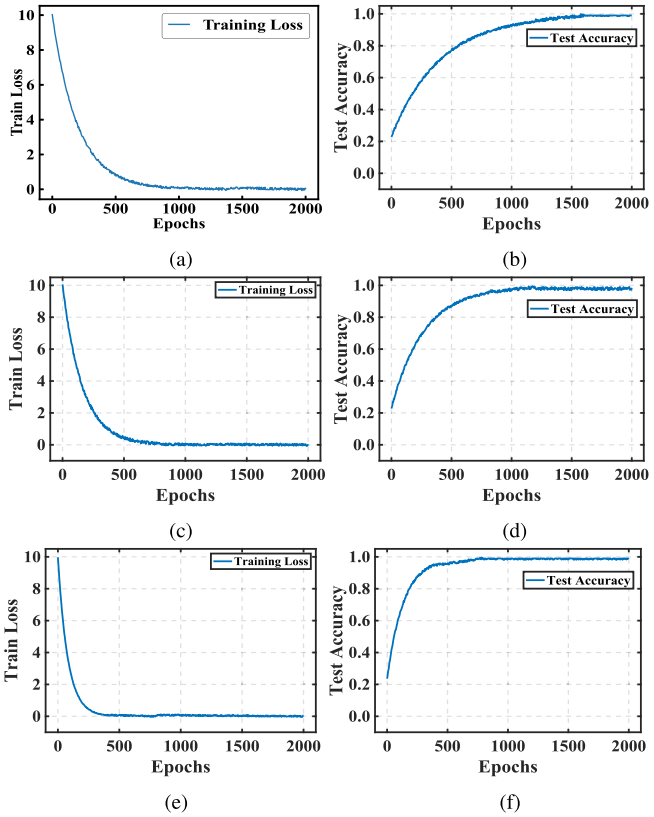


Fig. 8. During training and evaluation on three different dataset types, the training loss decreases, and the test accuracy of our HSTNet improves as the number of iterations increases. (a) and (b) Training loss and test accuracy on the real dataset. (c) and (d) Correspond to the generated dataset. (e) and (f) Results on the mixed dataset.

of the validation test set. For all three datasets, we discuss and analyze them with 2000 iterations as an example. For the real dataset, the training loss of our HSTNet remains stable around 1000 iterations, while the test accuracy remains stable around 1500 iterations. For the generated dataset, the training loss of our HSTNet remains stable after approximately 600 iterations, while the test accuracy remains stable after approximately 800 iterations. For the hybrid dataset, the training loss of our HSTNet remains stable around 450 iterations, while the test accuracy remains stable around 500 iterations. The study demonstrates that HSTNet achieves fast convergence and yields optimal classification results quickly as the amount of data increases.

E. Influence of Training Set Size

To systematically evaluate the influence of training data on model performance and verify the effectiveness of our proposed data generation method, we conducted a series of experiments on the influence of training set size. We compare the recall results of each classification method on three different types of datasets (real, generated, and mixed datasets) and with five different training-testing set division ratios (3:2, 7:3, 3:1, 4:1, and 5:1).

Fig. 9 shows the performance comparison of all models on (a) real datasets, (b) generated datasets, and (c) mixed datasets. Three key trends are evident from the graph. First, a universal rule is that the performance of all models steadily improves as the amount of training data increases. In all three subgraphs, when the ratio of the training-test set division increased from 3:2 to 5:1, each performance curve showed a distinct upward trend. This phenomenon indicates that increasing the number of training samples can effectively help the model learn more generalized features, thereby improving its performance on the test set. Second, our proposed HSTNet model consistently demonstrates significant superiority in comparison to individual model performance. Under all datasets and training ratios, HSTNet's performance curve (red solid line) consistently outperforms all compared models. For example, HSTNet with a 5:1 partitioning ratio achieves a recall of approximately 0.99, 0.983, and 0.995 on the real dataset, the generated dataset, and the mixed dataset, respectively. In contrast, the traditional 3DCNN model underperforms in most cases, which further highlights the effectiveness of our model architecture design.

The comparison results of each model on different datasets also validate the great value of our data enhancement strategy. As shown in Fig. 9(a) and (b), the performance of all models on the generated dataset is generally inferior to that on the real dataset, which indicates that there are still some domain differences between the generated data and the real data. When we mix the generated data with the real data, Fig. 9(c) shows that the performance of all models improves significantly, even surpassing the level achieved when using only the real data. Notably, HSTNet achieved the highest recall of 0.995 on the mixed dataset at a 5:1 ratio between the training and test sets, which is not only higher than its 0.99 on the real dataset but also significantly higher than the performance of any other model under any condition. This result powerfully demonstrates that the synthetic samples produced by the generative model we employed successfully enriched the feature space of the training set and improved the robustness and generalization ability of the model.

F. Ablation Studies

In our ablation study, we conduct comprehensive ablation experiments on HSTNet to validate the positive impact of individual modules on the classification of hyperspectral wheat images. Concretely, we conducted the following several key ablation experiments on the real dataset.

1) Our HSTNet without the data preprocessing (−w/o DPP).
2) Our HSTNet without the SCGAN (−w/o SCGAN).
3) Our HSTNet without the frequency domain multilayer perceptron (−w/o FDMP).
4) Our HSTNet without the CAC module (−w/o CACM).

It is noteworthy that based on extensive statistical analysis, our HSTNet demonstrates favorable performance when the window size w is set to 17×17 during the preprocessing stage.

In Table VI, we report the results of the individual ablation models for classifying 20 classes of the real hyperspectral wheat varieties. We can summarize the following results for the details of the data in Table VI. (Case 1) −w/o DPP: our HSTNet removed the DPP, which demonstrated a significant reduction in performance for hyperspectral wheat variety

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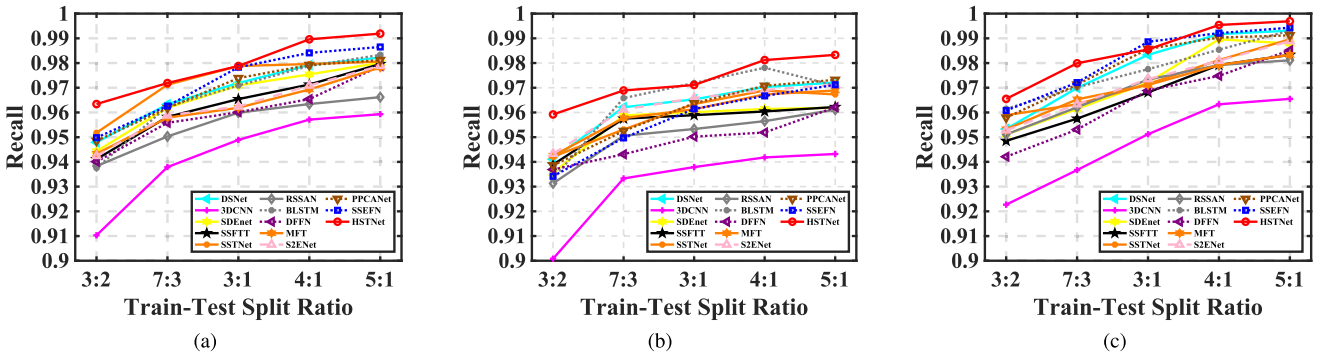


Fig. 9. Recall trend charts of each method were tested on real datasets, generated datasets, and mixed datasets with different ratios of training set and test set sizes. (a) Each method experiments on the real dataset. (b) Each method experiments on the generated dataset. (c) Each method experiments on the mixed dataset.

classification. The primary reason for this phenomenon is that the raw dataset contains more noise and exhibits larger spectral fluctuations, which have a greater impact on the classification. It can be observed that the DPP module has a significant effect on the whole HSTNet. (Case 2) −w/o SCGAN: Our HSTNet removed SSCGAN, which also led to a significant reduction in the classification accuracy of this model. The primary reason is that the ablation experiment was conducted without generating a dataset. In short, the insufficient number of training samples also impacted the model's classification performance. (Case 3) −w/o FDMP: Our HSTNet removes the FDMP, which slightly decreases the classification accuracy of this model. The FDMP maps the spatial information of the spectral data to the frequency information, thereby obtaining more detailed spectral information. (Case 4) −w/o CACM: our HSTNet removes CACM, which slightly reduces the model's classification accuracy. CACM is used to fully extract the weak features that may be lost after 2-D convolution and 3-D convolution, which also has a certain impact on the classification results. In Case 5, we separately analyzed the influence of window size, the number of MLP layers, and the combination strategy of 2DCNN and 3DCNN on the overall performance of our full HSTNet model, and the results in Table VI clearly indicate that HSTNet achieves its best classification accuracy when the window size is set to 17×17, the MLP consists of two layers, and both 2DCNN and 3DCNN are jointly utilized.

In summary, the quantitative results of this ablation study provide compelling evidence for the soundness of our HSTNet. The DPP, SSCGAN, FDMP, and CACM modules within HSTNet all make positive and critical contributions to the model's outstanding performance. Removing any single module leads to a significant decline in model performance, thereby validating the integrity and efficiency of our proposed HSTNet.

G. Complexity Analysis

Table VII presents computational efficiency metrics for all comparison methods on the real dataset. The best metric is marked in red, with the second-best in blue. HSTNet maintains approximately 39.95-M parameters, demonstrating strong competitiveness against methods such as SSFTT (34.55 M) and BLSTM (29.83 M). In terms of computational complexity, HSTNet exhibits faster training speeds at 192.84 million operations, demonstrating exceptional efficiency. These results confirm that HSTNet successfully balances model complexity with computational efficiency, rendering it particularly well-suited for practical HSI classification applications where both accuracy and speed are paramount.

TABLE VII
COMPARISON OF MODEL COMPLEXITY AND INFERENCE TIME

Methods	Real dataset			
	Param (M)	FLOPs (M)	Train (s)	Test (s)
DSNet [52]	15.87	26.01	9.61	2.70
3DCNN [53]	18.54	225.03	21.33	3.93
SDEnet [54]	51.66	59.58	13.94	3.59
RSSAN [50]	25.04	30.62	11.28	4.64
DFFN [55]	39.82	312.37	25.87	6.38
S^2ENet [11]	14.23	111.49	14.62	3.86
SSEFN [56]	37.83	150.74	15.71	6.45
SSFTT [4]	34.55	113.27	8.59	2.86
MFT [57]	31.83	118.74	10.78	3.77
BLSTM [25]	29.83	135.74	9.85	5.89
SSTNet [58]	31.45	226.66	18.91	5.28
PPCANet [59]	43.95	198.95	14.48	4.16
HSTNet	39.95	192.84	7.96	2.11

V. CONCLUSION

In this article, an HSTNet was proposed to address the challenge of fine-grained classification of wheat varieties using hyperspectral imaging. The network integrates an SCGAN for data augmentation and an STDNet for feature extraction and classification. SCGAN effectively enriches the training data by generating realistic hyperspectral samples, while STDNet leverages both 3-D and 2-D CNNs combined with an FFT and a CAM to extract and fuse spectral, spatial, and texture features. Furthermore, an MWHD containing 4800 samples from 20 wheat varieties was constructed to support the HSTNet training and evaluation. Experimental results on MWHD demonstrate that the proposed HSTNet outperforms several state-of-the-art methods, verifying its effectiveness in hyperspectral wheat classification tasks.

Although HSTNet shows strong performance, several limitations remain. The model still depends on high-quality spectral data and involves relatively high computational costs, which may hinder its application in real-time or edge-computing scenarios. In future work, efforts will be directed toward optimizing model efficiency, exploring self-supervised

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or few-shot learning techniques to reduce reliance on labeled data, and extending the framework to other crops and field conditions. These directions aim to enhance the practical applicability of HSTNet in large-scale smart agricultural systems.

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